

Assumptions Log

This document logs the assumptions made during the EMS Readiness Optimization project.

Data Assumptions

A1: Crash Data Representativeness

Assumption: Historical crash data (2013-2026) is representative of future crash patterns.

Rationale: Long time series provides stable estimates of spatial and temporal patterns.

Risk: Major changes in traffic patterns (e.g., congestion pricing) could alter future demand.

Mitigation: Include sensitivity analysis on demand levels.

A2: Location Accuracy

Assumption: Crash latitude/longitude coordinates are sufficiently accurate for demand estimation.

Rationale: NYPD geocoding process should provide reasonable accuracy for most incidents.

Risk: Some records may have missing or inaccurate coordinates.

Mitigation: Filter records with missing coordinates; aggregate to grid cells.

A3: Firehouse Availability

Assumption: All Manhattan firehouses are available as potential EMS staging locations.

Rationale: Study explores optimal staging locations; operational constraints addressed separately.

Risk: Some firehouses may not have space or infrastructure for EMS staging.

Mitigation: Document as limitation; can add constraints in future iterations.

Model Assumptions

A4: Travel Time Model

Assumption: Travel times can be approximated using Euclidean distance with a scaling factor or simplified network distances.

Rationale: Full routing adds complexity without fundamentally changing optimization insights.

Risk: May over/underestimate actual response times in some areas.

Mitigation: Sensitivity analysis on travel time multipliers.

A5: Single Unit Response

Assumption: Each crash incident requires exactly one EMS unit.

Rationale: Simplifies simulation while capturing core dispatch dynamics.

Risk: Major incidents may require multiple units.

Mitigation: Model captures most incidents; note as limitation.

A6: Homogeneous Service Time

Assumption: Service time (on-scene + transport) follows a single distribution.

Rationale: Data on actual service times not available; use literature estimates.

Risk: Service times may vary by incident severity.

Mitigation: Sensitivity analysis on service time parameters.

A7: Immediate Dispatch

Assumption: EMS units are dispatched immediately upon incident occurrence.

Rationale: Focuses on travel time rather than dispatch delay.

Risk: Real systems have dispatch processing time.

Mitigation: Can add dispatch delay parameter if needed.

Geographic Assumptions

A8: Manhattan Boundary

Assumption: Incidents and firehouses are classified by whether they fall within Manhattan borough boundary.

Rationale: Standard geographic classification for NYC.

Risk: Boundary edges may have some ambiguity.

Mitigation: Use official borough boundary polygons.

A9: CBD Definition

Assumption: CBD is defined as the MTA Congestion Relief Zone boundary.

Rationale: Provides consistent, official boundary for the high-density core.

Risk: Other CBD definitions exist; results depend on boundary choice.

Mitigation: Document specific boundary used; can test alternatives.

Operational Assumptions

A10: Firehouse Staging Capacity (Updated v1.3.0)

Assumption: ~Any number of EMS units can stage at a single firehouse.~ **Updated:** Maximum 2 EMS units per firehouse (default capacity constraint).

Rationale: Full capacity sensitivity analysis (cap 1-5) shows that cap=2 is operationally realistic based on typical FDNY firehouse infrastructure, and matches or improves performance compared to cap=5 at $K \leq 40$. At $K=20$, capacity never binds (all policies allocate ≤ 1 unit/firehouse). At $K=40$, cap=2 forces wider geographic dispersion (29 vs 24 firehouses for P2).

Risk: Some firehouses may accommodate more or fewer units depending on physical infrastructure.

Mitigation: Capacity sensitivity analysis (cap 1-5) quantifies performance impact. Results are robust across all tested capacity values.

A11: No Pre-emption

Assumption: Once an EMS unit is assigned to an incident, it completes service before taking new calls.

Rationale: Standard EMS operational policy.

Risk: Pre-emption for high-priority calls does occur.

Mitigation: Aligns with most common practice; note limitation.

A12: Manhattan Distance Approximation

Assumption: Manhattan (taxicab) distance provides an upper-bound approximation of actual road distance in a grid-based network; Haversine provides a lower bound.

Rationale: Manhattan's street grid is largely rectangular; true road distance falls between Haversine and Manhattan distance.

Risk: Diagonal avenues and non-grid areas (e.g., lower Manhattan) deviate from both approximations.

Mitigation: Compared both metrics side-by-side; P2 allocation is robust to metric choice (mean RT differs by <0.01 min in simulation).

A13: CBD-Focused Demand Weighting

Assumption: Applying a 3× demand weight to CBD precincts in the CBD-focused model is a reasonable multiplier to test CBD prioritisation.

Rationale: The 3× factor substantially up-weights CBD demand (from 56% to ~77% effective weight) without completely ignoring non-CBD.

Risk: A different multiplier (e.g., 2× or 5×) might yield different conclusions.

Mitigation: Results show that even with 3× weight, CBD RT does not improve (2.50 vs. 2.47 min), while non-CBD RT degrades dramatically (+159%). This confirms that the Manhattan-wide P2 is robust and any CBD-focused up-weighting is counterproductive.

A14: Spatially-Stratified Baseline (P0)

Assumption: The baseline uniform allocation policy (P0) distributes units across firehouses using latitude-based spatial stratification rather than arbitrary index-based ordering.

Rationale: Latitude-based stratification divides Manhattan into equal-width latitude bands and rounds units across bands, ensuring even north-south geographic coverage regardless of firehouse ordering in the dataset. This produces a fair, geographically representative baseline.

Risk: Latitude-only stratification does not account for east-west variation or local demand density.

Mitigation: Manhattan's elongated north-south geometry makes latitude the dominant spatial axis. P0 is a baseline comparator; optimized policies (P1, P2) already account for demand. Simulation confirms P0 achieves 3.17 min mean RT at K=20 with 99.6% 8-minute coverage.

A15: P0 Nomenclature (Internal Reference)

Assumption: The project uses a single, standardized P0 definition (spatially-stratified allocation) throughout all public-facing documents. An earlier index-based implementation (deprecated) exists in code for backward compatibility but is not referenced in the technical report.

Rationale: The original index-based P0 was inadvertently biased toward CBD firehouses due to database ordering. This produced misleadingly poor baseline performance (8.08 min mean RT at K=20). The spatially-stratified P0 corrects this bias and provides a fairer comparison point for P1 and P2. See `docs/decisions_log.md` (DEC-012) and `docs/nomenclature_migration.md` for the full evolution history.

Risk: Readers of older project artifacts may encounter inconsistent P0 definitions.

Mitigation: Internal migration guide (`docs/nomenclature_migration.md`) explains the evolution. Code retains `uniform_allocation()` with `DeprecationWarning`.

Assumption Review Schedule

Assumption	Review Date	Reviewer	Status
All	Initial	Project Lead	Documented

Last Updated: March 15, 2026